SUBJECT: FINAL RED TEAM AUDIT & EXECUTIVE BOM DIRECTIVE (MAYA EDGE-OS) (10/10/2026)

Prices are based on market price + / -

To: Bharath CD & Balaraj R (Engineering Team) **From:** Girimallikarjuna SMD, Managing Director, Novabion Pvt. Ltd.

Your V5 component list was reviewed. I did not share the locked V4 list with you over the last 48 hours, so you were operating on assumptions based on standard lab prototyping.

Today, we are locking the exact, field-ready **Phase 1 MVP Unit BOM**, separating out your lab tooling budget, and authorizing procurement.

In rural Chitradurga, hardware does not fail because of bad Python code. It fails because of 280V voltage spikes, 35°C ambient heat, fine limestone dust (*Sunnakattu*), and high-pressure chemical bleaching powder washes. We are building an industrial IoT enclosure that survives these conditions, while keeping the manufacturing cost under the psychological ₹50,000 retail barrier.

Here is the exact component breakdown and the Red Team reality of why certain components were rejected or added.

1. The Red Team Component Corrections

A. Core Compute & AI (The 4GB Pivot): We are dropping the 8GB Raspberry Pi 5 to the 4GB variant. MAYA Edge-OS is a headless system. Because we are pairing it with the Hailo-8L M.2 AI HAT, the Hailo NPU handles 100% of the YOLO vision and matrix mathematics. The Pi's RAM is strictly used for routing data and saving text to SQLite. 4GB is more than sufficient; paying an extra ₹3,000 for 8GB is a waste of capital. *Note:* We are keeping the **13 TOPS Hailo-8L**, not the 26 TOPS Hailo-8. 13 TOPS is massive overkill for tracking slow-moving silkworm biological events at 1–5 frames per second.

B. The Industrial Power Failsafe: We have **deleted** the standard Pi USB-C wall charger and the consumer 5V Power Bank. A consumer lithium power bank doing 24/7 pass-through charging in a hot shed will swell and create a fire hazard. We have replaced these with an **Industrial DIN-Rail SMPS (5V/5A)** to filter the shed's dirty 150V-280V grid power, paired with a dedicated **Raspberry Pi UPS Battery HAT (18650 cells)** for true zero-millisecond power failover.

C. The Vision Stack (Camera & Screen): We have **deleted** the bare Arducam USB camera. USB cables suffer severe signal degradation past 3 meters and cannot cross a 40-foot commercial shed. We are reverting to the **TP-Link Tapo C310 IP66 Camera**, which connects over the local

network (RTSP) and survives mister humidity. We will **keep** the 5-inch HDMI Touchscreen you requested, but with a strict installation rule: it must be mounted on the **outside** of our sealed IP65 box, acting as an external "Kiosk UI" for the farmer. If the screen is exposed to the inside of the shed, bleaching powder will destroy it in a week.

D. The Missing Chemical Defense: We have added a **Sintered PTFE (Teflon) Filter Cap** to the Sensirion SHT40 Temp/Humidity sensor. Farmers spray highly corrosive 'Right Bleaching' powder and chemical solutions between batches. The PTFE cap allows air to pass through to the sensor but repels liquid chemicals and dust, preventing total sensor failure.

2. The Final Locked Unit BOM (MAYA Edge-OS MVP)

You are cleared to procure exactly these parts for the first unit. Source the Pi 5 exclusively from official B2B channels (Element14 / Robu / Silverline) to bypass the ₹20k Amazon scalper markup.

Sl. No	Component Name	Qty	Cost (INR)	Primary Use Case in Architecture
1	**Raspberry Pi 5 (4GB RAM) - Bare Board**	1	₹10,244	Core Edge OS, network routing, and local SQLite data logging.
2	**Raspberry Pi AI HAT+ 13 TOPS (Hailo-8L)**	1	₹7,469	Offloads 100% of the YOLO11n AI vision processing.
3	**Official Pi 5 Active Cooler**	1	₹531	Prevents thermal throttling during 24/7 edge computation.
4	**SanDisk Ultra microSDXC 128GB (A1)**	1	₹2,649	High-endurance flash for OS and offline event queue.
5	**Industrial DIN-Rail SMPS (5V/5A, 25W)**	1	₹800	Cleans fluctuating 150V–280V grid power to protect the Pi motherboard.
6	**Raspberry Pi UPS Battery HAT + 18650 Cells**	1	₹2,200	Provides zero-millisecond failover during rural power blackouts.
7	**TP-Link Tapo C310 V2 Outdoor Camera (IP66)**	1	₹2,999	Local RTSP network video stream; immune to shed misters.
8	**Daylight White LED Strip (12V, Waterproof)**	1	₹517	Pulses for 10 seconds at night for RGB snapshots (AI fails on IR).

9	**Sensirion SHT40 Temp & Humidity Breakout**	1	₹698	Captures ultra-precise micro-climate data via I2C.
10	**Sintered PTFE (Teflon) Sensor Protective Cap**	1	₹250	Blocks corrosive bleaching powder and lime dust from melting the SHT40.
11	**Magnetic Float Level Switch (Vertical, 2-Wire)**	1	₹122	Prevents the mister pump from running dry and causing a motor fire.
12	**Waveshare 6-Channel Isolated Relay (SSR)**	1	₹4,400	Optically isolated 220V AC load switching for shed fans and misters.
13	**Dual-Value Fan Motor Capacitor (2.5µF/4µF)**	1	₹80	Enables "Gentle Breeze" 2-speed fan control without an expensive VFD.
14	**5-Inch HDMI LCD Kiosk Display (800x480)**	1	₹2,359	External UI kiosk mounted on the outside of the box/shed door.
15	**Micro HDMI to HDMI Cable**	1	₹189	Display interconnect.
16	**IP65 Weatherproof Industrial Enclosure**	1	₹1,800	Fully seals the internal electronics against moisture and dust.
17	**Aviation Connectors & Industrial Wiring**	1	₹1,000	GX16 quick-disconnects for rapid plug-and-play unit swapping.
18	**Passive Electronics Interface Kit**	1	₹480	Toggle switches, buzzer, LEDs, resistors, Dupont wires, and prototype PCBs.
--	**TOTAL EXACT MANUFACTURING COST**	--	**₹38,787**	**Final Authorized Unit Budget**

(Note: The Poultry I2S Digital Microphone Array (₹3,289) has been removed from this Sericulture-specific BOM and will be procured separately in Phase 1.5).

3. One-Time R&D Lab Tooling Authorization

We do not ship screwdrivers and multimeters to the farmer. These are one-time capital expenses for our Chitradurga engineering lab. I am authorizing a separate, one-time procurement of **₹2,069** to establish your workbench:

- Digital Multimeter: ₹799
- Utility Knife & Wire Cutters: ₹629
- Screwdriver Set & Pliers: ₹333
- Soldering Iron, Paste & Wire: ₹308

Immediate Execution Steps:

1. **Checkout:** Go through the document and finalize, we will order today.
2. **Software Parallel:** While the hardware ships, begin flashing the OS image, write the SQLite schemas, and initialize the CVAT workspace.

We are officially cleared for the Phase 1 build. Execute.